



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

ETH Zurich Career Seed Award Proposal

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1. Project Identifiers

Corresponding Applicant:
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Co-Applicant, if applicable:

Title of project: **Large-Scale Web Archives as a Source for Media Research:
A Proof-of-Concept on Hungarian Media Capture**

Submitted for deadline: 1 March 2026

Disclosure on the use of generative artificial intelligence

Claude Chat (Sonnet 4.6) was used to improve formulations of this grant proposal.

2. Achievements at ETH Zurich as postdoctoral researcher

2.1 Description of the applicant's ongoing work in the host lab

My research agenda focuses on the role of media in contemporary democracies. It studies this question by combining cutting-edge machine learning techniques with experimental and causal inference methodology. As Postdoctoral Researcher in the Public Policy Group at ETH Zurich, I have developed a comprehensive data pipeline that automatically processes hundreds of thousands of online comments from multiple media partners each week. This infrastructure serves as foundation for our field experiments and the *Public Discourse Indicator*, a dashboard monitoring online discourse in Switzerland. The system has processed over 50 million comments to date. One crucial aspect of this work was the development of text classifiers for the measurement of different debate quality aspects, which required the supervision and training of multiple research assistants to achieve the highest data quality and the fine-tuning of state-of-the-art transformer models. The data and classifiers were used in several major field experiments testing scalable interventions to improve online debate. These experiments, implemented in collaboration with our industry partners, manipulate platform design features in live commenting environments, allowing causal identification of what drives hostility and deliberation online. Our studies show that algorithmically preventing the publication of toxic content reduces the toxicity of submitted content by 23%, suggesting online norms are strongly shaped by perceptions of others' behavior. In a separate study, visible moderator participation alone reduced toxic submissions by 33%. Several working papers based on this work have been presented at conferences in the past year and are currently being prepared for submission.

2.2 Scientific achievements and research output during the applicant's stay at ETH Zurich as postdoctoral researcher

In addition to the significant work within the research group, I developed my own research agenda highlighting the surprising resilience of public opinion: My first paper, published in *Political Communication*, analyzed 2.5 million German news articles using transformer models and panel data from 15,000+ individuals to show that a dramatic shift in immigration framing had no detectable consequences for immigration attitudes. This work received the International Communication Association's Kaid-Sanders Best Paper Award and the Elizabeth Meehan Prize. My dissertation monograph confirms this finding for a variety of immigration-related events. A second paper, published in *Political Behavior*, experimentally shows that while weak attitudes respond to news framing, these changes do not translate into voting intentions. Another working paper on vaccination rates and voting behaviour during the Covid pandemic is currently in development.

Responding to developments in liberal democracies around the world, I have expanded my research on the media's role in the erosion of democratic institutions. In collaboration with Dr. Markus Kollberg (Humboldt University Berlin), I conducted a survey experiment to understand whether informing citizens about liberal democratic norms and the process of their erosion makes them more inclined to resist democratic backsliding. We are currently designing a larger, multi-country follow-up study.

I have presented my research at several invited talks at the *University of Aarhus*, *Central European University Vienna*, *University College Dublin*, and *20 Minuten*, and taught an intensive workshop on large language models for social science research to PhD students as part of the CUSO doctoral program. I also spent a month at the Centre for Media Pluralism and Media Freedom (CMPF) at the European University Institute to develop my research agenda on the role of media pluralism in contemporary democracies.

Lastly, I have substantially contributed to the development of text-as-data research in the social sciences. I founded and co-organize the Text-as-Data (TADA) initiative, now in its fifth year, which fosters crossdisciplinary collaboration through regular seminars and an active community of over 750 members. The series has hosted 50+ talks by leading NLP researchers from Europe, North America, and Asia.

3. Scientific Information

3.1 Rationale for the proposed project and state of research

Theorists of democracy have often underscored the relevance of an unconstrained and diverse information environment for the functioning of liberal democracy, be it as a ‘marketplace of ideas’ promoting better policies, or as a public sphere to deliberate about policy and politics [1]. However, research on this fundamental link is scarce, largely due to the absence of comparative data on media pluralism. Major comparative research projects have primarily focused on the monitoring of media policy [2], media consumption [3], or media freedom [4]. To date, not a single dataset is concerned with the ideological diversity of media markets. This lack of data inhibits our understanding of the contemporary media landscape and its relationship with democratic stability. A major reason for this gap is the practical difficulty of obtaining media content data at scale. Media research has traditionally relied on outlet-specific web scrapers, which break whenever a website changes its structure, requiring constant maintenance. In addition, data on the historical content of websites can usually not be collected with this approach. In the proposed project, I want to provide a proof-of-concept for a largely untapped alternative: large-scale web archives, most notably CommonCrawl's CC and CC-NEWS corpora [5], which has indexed billions of web pages since 2008. CommonCrawl provides freely available, historically archived full-text content from thousands of outlets across languages, offering a stable foundation for systematic, longitudinal measurement of media content. Despite its potential, and wide-spread application in the training of large language models, this data source has seen no use in political science or communication research. Beyond providing a proof-of concept for the use of the CommonCrawl corpus for media research, this project exploits CommonCrawl to develop a novel content-based measure of media pluralism and applies it to the paradigmatic case of media capture in Europe: Hungary. Since 2010, the Orbán government systematically captured the Hungarian media through a sequence of interventions, such as a partisan media law, the takeover of independent outlets, the targeting of opposition outlets, and the unprecedented consolidation of a large number of outlets into a single government-aligned foundation. Each event is extensively documented qualitatively, yet no study has applied causal inference methods to measure their impact on content diversity. The Hungarian case thus serves as an ideal proof of concept for the measurement approach using data from web archives.

Resources

- [1] [Habermas, J. (1974). *The public sphere: An encyclopedia article*. New German Critique, 3, 49–55.
- [2] Blagojev, T., Bleyer-Simon, K., Brogi, E., et al. (2025). *Monitoring media pluralism in the EU*. European University Institute.
- [3] Newman, N., Arguedas, A. R., Robertson, et al. (2025). *Digital news report 2025*. Reuters Institute for the Study of Journalism.
- [4] Neundorff, A., Nazrullaeva, E., Northmore-Ball, K., et al. (2023). *Varieties of indoctrination*. (V-Dem Working Paper No. 136).
- [5] Nagel, S. (2016, October 4). *News dataset available*. Common Crawl. <https://commoncrawl.org/blog/news-dataset-available>

3.2 Research plan

Objective. This project provides a proof-of-concept to use large web crawls as a source of media data at scale and develops a novel, content-based measure of media pluralism. An application in Hungary causally identifies the impact of government interventions on media content diversity in Hungary.

Data collection. I will extract the full text of all articles published by major Hungarian news outlets from the Common Crawl web archive. The primary data source is the Common Crawl News corpus (CC-NEWS), which has collected news content automatically via RSS feeds and sitemaps on a daily basis since September 2016 (Nagel 2016), providing comprehensive and reliable coverage of Hungarian news domains from that date onward. For the period before 2016, I will draw on the main Common Crawl corpus (CC-MAIN), which has archived web content since 2008, with regular, indexed monthly crawls available from 2013. While CC-MAIN does not target news content specifically, its broad web crawls capture news domains alongside other websites, and its indexation allows targeted retrieval of specific domains. The population of outlets will be defined using consumption data from the Reuters Digital News Report. The resulting corpus will contain the full text of articles from approximately 30–40 Hungarian outlets over eleven years. The integrity of the data will be assessed based on various checks of the distribution of content features (such as text length), data availability over time, and the face validity of randomly selected content.

Measurement. To estimate the political positioning of outlets over time, I define four dimensions of outlet positioning: stance towards the government, support for liberal democracy, economic left-right orientation, and cultural policy orientation. Each dimension is operationalized at the article level — for instance, whether an article criticizes the current government. To generate training data for the classifiers, I draw on recent advances in LLM-based synthetic annotation. For a random sample of

15,000 articles, a state-of-the-art large language model (such as Claude Sonnet 4.6) annotates articles on each dimension. These annotations serve as training data for a multilingual natural-language inference (NLI) classifier (Laurer et al. 2024), fine-tuned to predict article-level positions for the full corpus. The classifier is validated against a held-out sample of approximately 2,500 articles hand-coded by a Hungarian-speaking research assistant, and any systematic measurement error is adjusted using design-based supervised learning to generate unbiased outlet estimates (Egami et al. 2023). Outlet-year positions on each dimension are computed by aggregating article-level scores. These outlet-level estimates are then combined into a market-level media pluralism index, weighted by audience reach from the Reuters Digital News Report, capturing the effective diversity of the Hungarian information environment over time. As a final validation step, I will collect pairwise article comparisons and outlet positions in a small representative survey to validate the model predictions and assess the public perceptions of outlets. This triangulation across human perceptions, LLM judgments, and classifier output provides a robust assessment of measurement validity, while the perceptions of outlet positions provide interesting data about public media perceptions.

Empirical Application. I employ event study designs around major government interventions in the Hungarian media landscape. The CC-NEWS corpus (from 2016) provides reliable data for the three most dramatic events: the abrupt closure of opposition daily Népszabadság (October 2016), the consolidation of a large number of outlets into the KESMA foundation (November 2018), and the takeover of Index.hu (July 2020). If an assessment of the coverage in the CC-MAIN corpus confirms sufficient data availability for earlier periods, I will extend the analysis to the capture of Origo.hu following the dismissal of its editor-in-chief (2014) and, data permitting, the establishment of the partisan media authority NMHH (2010/2011). The impact of each intervention on the ideological composition of the news market will be estimated using difference-in-difference and synthetic control designs, comparing affected outlets with unaffected outlets. I will then estimate the downstream consequences for media pluralism. This will provide crucial evidence on which changes in national media markets are most sensitive and should serve as red lines for liberal governments.

3.3 Expected output and significance / expected impact

The project will produce two papers. The first introduces Common Crawl as a data source for comparative media research, demonstrating the data retrieval pipeline, and assesses coverage across outlets, languages, and time periods on a meta-level, as well as presenting the Hungarian corpus as an application. This methodological contribution is targeted at *Political Communication* or *Political Analysis*. The second paper develops the content-based media pluralism measure, applies it to the Hungarian case, and presents causal evidence on how government interventions erode media pluralism. This paper is targeted at a general political science outlet such as the *American Political Science Review* or the *Journal of Politics*. More generally, the methodological approach developed as part of this project will provide the foundation to prepare a larger grant to measure media pluralism comparatively across many countries and to investigate its role in the stability and erosion of contemporary democracies.

3.4 Use of funding

Hungarian-speaking research assistant (~400 hrs at CHF 32/hr)	13,000 CHF
Processing Infrastructure (e.g. ~500h AWS c5.12xlarge)	1,500 CHF
Storage (Postgres DB and AWS S3)	1,500 CHF
Generation of training labels with LLM (100k labels on 6 dimensions)	1,000 CHF
Model training (~300h AWS p3.8xlarge)	3,000 CHF
Inference (~300h AWS g4dn.12xlarge)	1,500 CHF
Computational infrastructure total	8,500 CHF
Representative survey in HU for validation of outlet estimates and additional annotations for inter-coder reliability (~1000 respondents)	5,000 CHF
Conference Travel and Collaboration	3,500 CHF
TOTAL	30,000 CHF

The largest expense is the Hungarian-speaking RA, who is essential for developing and validating the NLP-based measures against human judgment — a critical step for establishing measurement credibility in a new language context. Computational costs reflect the scale of filtering and processing eleven years of web archive data across dozens of outlets, and the inference and training of LLMs on capable GPUs. The survey enables additional validation and a contrast of the ideological estimates with public perceptions of the outlets. Travel and conference costs ensure engagement with area specialists (at the Centre for Media Pluralism and Media Freedom) and dissemination of results.